Geospatial Analysis Portfolio

Katherine M. Yut

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Cover Letter

To Gianna, Nicole, and Jade,

I am writing to express my strong interest in a position at Terraformation and outline my qualifications, which firmly position me as an outstanding potential team member. With a background in geospaital data wrangling, data engineering pipeline orchestration, and the art of data visualization, I possess strong problem-solving abilities coupled with a knack for coding, making me an ideal candidate for scaling data-driven endeavors.

Rather than give an overview of each of my previous roles, I would like to focus on the growth I've achieved in my current role with Grassroots Carbon. Starting as a Carbon Technical Manager (a non-specific title), my work was underpinned by automating geospatial flows using R and Python. In that time, I used many of the skills I developed throughout college and in the following years. However, within the last year, I have advanced to a Data Engineer, acquiring skills such as writing advanced Python, deploying pipelines via Docker containerization, managing quality checks using the Pydantic framework and a custom-built Streamlit dashboard, maintaining documentation in GitHub, and more.

I am highly motivated to learn quickly and tackle new problems often. I am a pleasure to work with and am happy to provide more informal references who can support this claim. I am fully capable of working independently when necessary, but love to collaborate with others–I believe that working together has synergistic effects where the sum is greater than its parts. Providing support to and being supported by one another enhances effective communication, allowing the startup machine to run efficiently.

My educational background, consisting of a Master of Arts in Economics and Big Data, a Bachelor of Arts in Economics and Big Data, and a Bachelor of Science in Geographic Information Science, demonstrates my curiosity, rapid learning capability, and unwavering commitment to acquiring expertise across domains. My current skill set includes Python, GitHub, Docker, CI/CD, R, ESRI ArcGIS, QGIS, Google Earth Engine, and SQL.

Driven by a genuine passion for solving interesting problems with kind people, I am eager to contribute my insights, skills, and efficiency to the team. Thank you for considering my candidacy; I hope we can find a mutually beneficial alignment for a role within Terraformation.

Best,

Katherine (Katy) Yut

Katherine M. Yut

Geospatial Data Engineer

Expert in geospatial analysis, statistics, and programming using R and Python. Passionate about data visualization. Seeking to solve interesting environmental problems. Meticulous, intrinsically motivated, and a quick learner.

Work History

***Grassroots Carbon**, San Antonio, TX (hybrid)*
Data Engineer 2025-08 – current
Building ETL/ELT data pipelines using Python, GitHub, and Docker containerization.
Skills: pydantic, Streamlit, cron, GitHub Actions, CI/CD.

***Grassroots Carbon**, San Antonio, TX (hybrid)*
Carbon Technical Manager 2024-08 – 2025-08
Automated rancher eligibility screening processes using R shiny and Python, provided geospatial data analysis using QGIS, and supported the navigation of carbon standards.

***University of Texas, Bureau of Economic Geology**, Austin, TX*
GIS & Energy Data Analyst 2022-05 – 2024-07
Leveraged R to clean and manipulate data tables, solve problems, and automate processes; utilized ArcMap to aid in oil & gas research publications; and published research findings as papers, posters, and presentations.

Education

Master of Arts: Economics and Big Data
Bachelor of Arts: Economics and Big Data
Bachelor of Science: Geographic Information Science
University of Oklahoma (Norman, OK)
2016-08 – 2021-05, *Summa cum laude*

Research

KM Smye, **K Yut**, RC Reedy, BR Scanlon, JP Nicot, P Hennings (2024). Challenges with managing unconventional water production and disposal in the Permian Basin. AAPG Bulletin 108 (12), 2215-2240.

Peng, S., Maraggi, LMR., Bhattacharya, S., **Yut, K**., McMahon, T., Haddad, M. (2023). Feasibility of CO2 storage in depleted unconventional oil and gas reservoirs: Capacity, microscale mechanism, injectivity, fault stability, and monitoring. URTeC 13–15 June 2023.

M Bull, T Francis, **K Yut**, K Weber, J Spruce, C Jarnevich. West Virginia Ecological Forecasting. NASA DEVELOP National Program, 2019.

Kelley, S., **Yut, K**., Kulkarni, R., & Gaffin, D. (2019). Avoidance of rosemary oil by scorpions. Journal of Arachnology.

Contact

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Skills

Expert

RStudio, R, R shiny:
Writing functions, making packages, documenting code, creating interactive html servers.

ESRI ArcGIS Pro, QGIS:
Geospatial analyses, coordinate system management, map-making.

Creating timely deliverables and communicating effectively

Curiosity, organization, multitasking, and focus

Advanced

Python

GitHub

Big Data statistical processing, analysis, and modeling

Intermediate

SQL

Docker containerization

Google Earth Engine

ENVI

References

The following individuals are happy to vouch for my character and data analysis capabilities:

1. Chip Thompson Grassroots Carbon | Chief Financial Officer
   chip@buildgrassroots.com | (210) 687-3594

2. Katie Smye
   Bureau of Economic Geology | Research Associate
   katie.smye@beg.utexas.edu | (512)-471-6775

3. Brent Sams
   E. & J. Gallo Winery | Research Scientist
   sams.brents@gmail.com | (209) 568-9317

Sample Work

Grassroots Carbon

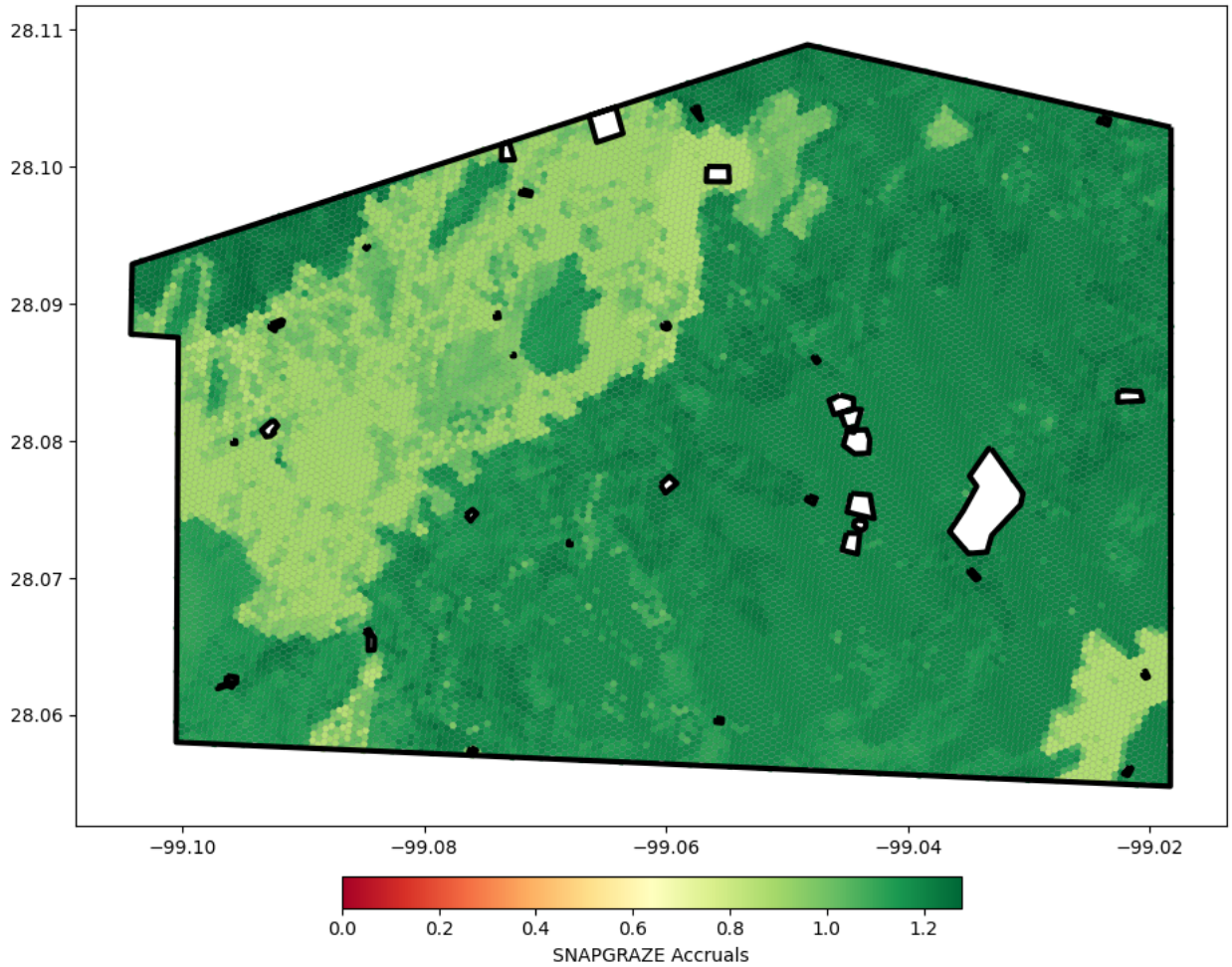

Figure 1: Heatmap of SNAPGRAZE model outputs, showing SOC accrual potential on a ranch in Texas. Created in Python using matplotlib and uber H3 hexagons.

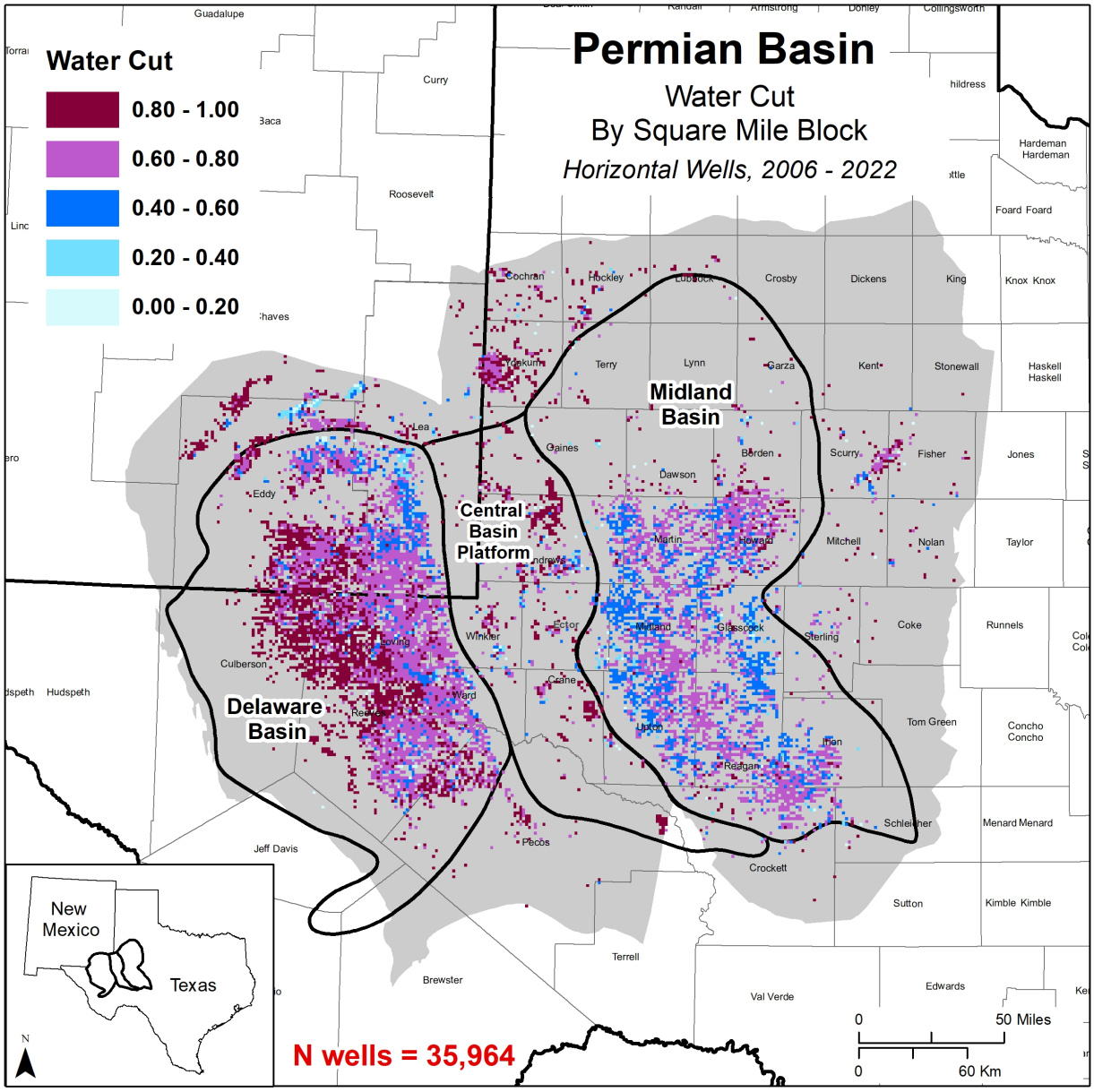

Figure 2: Map of Water Cut (Produced Water / Produced Oil + Produced Water) for all horizontal wells in the Permian Basin, aggregated on square mile blocks. Created in ArcMap using data from IHS Enerdeq.